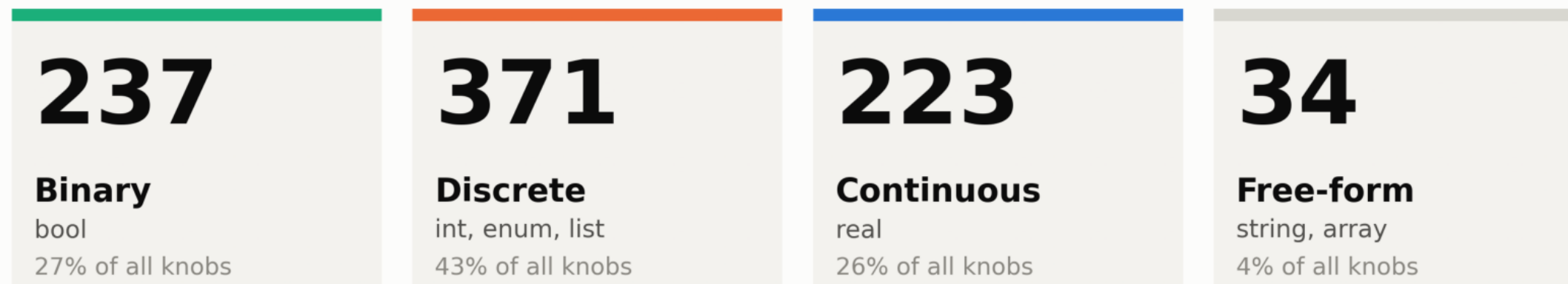


The solver parameter space

Every knob the nonlinear solver stack exposes, counted by the type of value it takes.

865 tuneable knobs

831 of them take a value worth searching. 57 come from the calling code's own input file; the rest are the solver library's.



One knob per row of the inventory. Counts are exact; nothing on this figure is estimated. 785 of the 865 are confirmed present in the library build actually linked: 728 solver-library names found in it plus 57 from the calling code. The other 80 are mostly preconditioners this build does not compile, so absence here is weaker evidence than presence.

Hermes-3 Solver Optimisation: Architecture & Flow

The test4 Campaign: multi-tier repositories, rung-based screening, noise-aware evaluation, and durable findings

CAMPAIGN: test4-jacobian

1. Three-Tier Architecture

Clear separation of framework, evidence, and transient cases

hermes-solver-opt (Public framework)

- Automation tools: runner, slice tools, validators
- Living requirements: R1-R38 with stable IDs
- Task tracking: beads issue tracker (Dolt DB)
- Architectural findings: campaigns/test4-jacobian/findings.md

solver-opt-store (Private evidence)

- Central immutable index: one row per run
- Run bundles: step logs, residual shares, traces
- Study manifests: S01, S02, S03 (.toml budgets)
- Standardised reporting: 10-section analysis, auto-generated briefs, HTML and PDF reports

\$data/cases/ (Transient storage)

- Simulation working directories & seeds
- Raw BOUT++ dumps (~1 GB per run)
- Diagnostic extraction pipeline before deletion
- Dumps pruned promptly to bound disk usage

2. test4 Evaluation Ladder & Noise Gate

Cheap slices filter candidates; only survivors pay for full simulation

Target Problem: test4 transient (2-6 ms holds 92% of full simulation time)

Rung 0 Screen Many: short slices (3.0-3.2 ms, 4.0-4.1 ms)

- Cost: ~200 s per run (seeded from saved reference states)
- Purpose: screen 20+ candidate solver settings cheaply
- Output: eliminates 80%+ of ineffective configurations fast

Rung 1 Test Survivors: intermediate slices (3.0-4.0 ms, 4.0-5.0 ms)

- Cost: ~1000 s per run (~5x Rung 0)
- Purpose: verify that Rung 0 speedups persist over longer horizons
- Cross-rung check: confirm rank order agreement before promotion

Promote
Winners

Rung 2 Confirm Finalists: full simulation (0.0-100.0 ms)

- Cost: 2500 s to 17 hours (full production turnaround)
- Purpose: prove real scientific throughput gain across full physics

Confirm
Finalists

Noise & Repeat Control

- 3 to 10 repeats per cell: measures numerical path noise on the 2.65 ms cliff
- Strict noise floor: 15% bound on Rung 0. No claim without clearing spread!

3. Studies & Decision Protocol

Hypothesis testing, gate criteria, and durable learning

Campaign Studies & Investigations

S01 Jacobian Lag Screening (22 settings, 167 runs)

- Overtaken production dogma: lag 1 beats lag 3 by 1.3x-2.6x
- Identified timestep controller gains as the next major lever

S02 Repeat Noise & Path Bifurcation

- Quantified run-to-run spread on the stiff 2.65 ms solver cliff

S03 Layered Options & Controller Gains (898 runs)

- Explored kl/kP gain plane; layered lag 1 + tuned PID gains
- Achieved up to 26x transient speedup and 2.5x full-run win!

I01 / I02 Targeted Diagnostic Investigations

- I01: core-edge residual; I02: direct solver round-off divergence

Promotion Decision Gate

- Physics verification: separatrix & target profiles match < 0.04%
 - Repeat clearance: speedup strictly exceeds declared noise bound
 - Cross-rung consistency: Rung 0 speedup confirmed on Rung 1
- Action: PROMOTE to next rung, REJECT, or QUARANTINE

Durable Findings Log

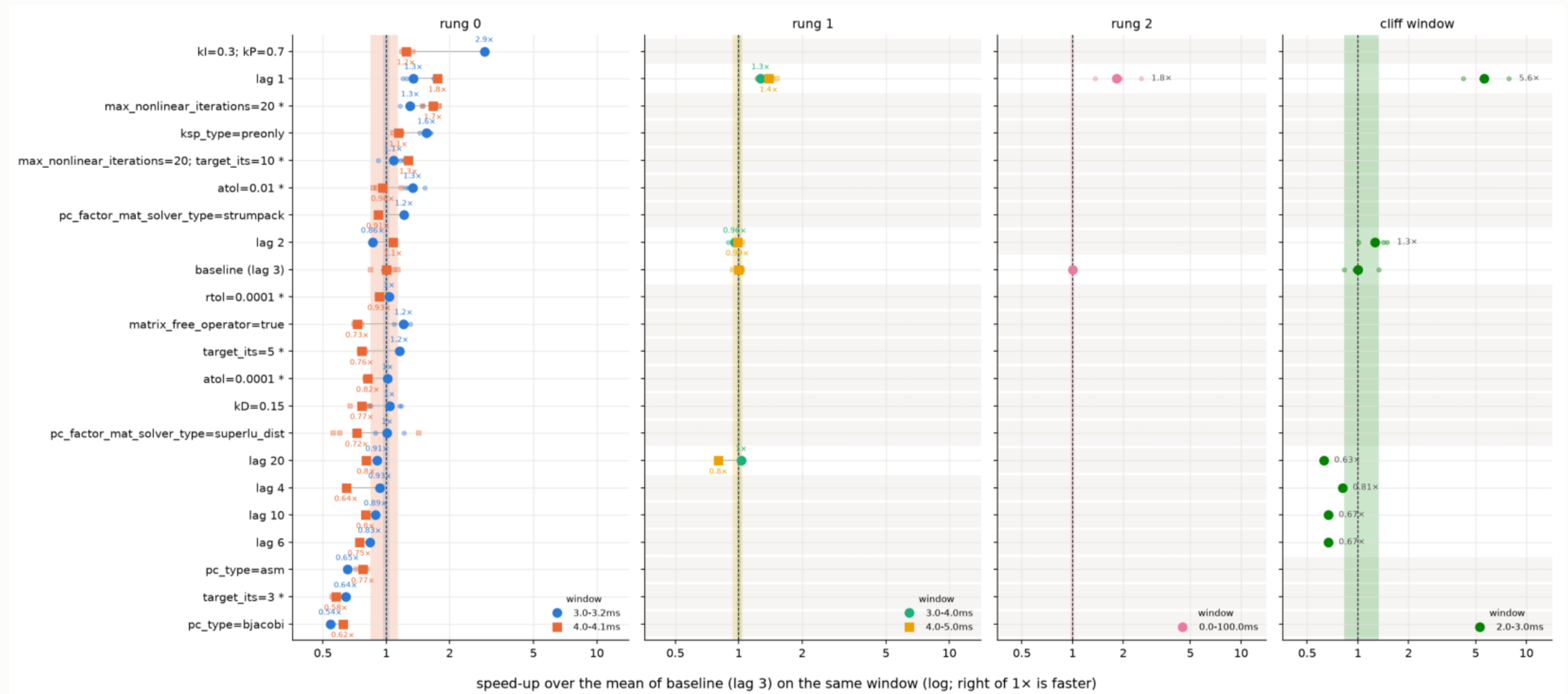
- Structured records: date, status, scope, evidence, actionable rule
- Recorded in campaigns/test4-jacobian/findings.md and root findings.md
- Graduated into production solver recipe defaults and requirements

Graduated findings update framework requirements & production defaults

HEADLINE ACHIEVEMENT: 865-knob space → S01 lag sweep (1.3x-2.6x) → S03 layered gains (up to 26x transient / 2.5x full run) → Zero physics loss (<0.04%)

Study S01: Jacobian Lag Screening

Overturning production dogma: updating the Jacobian every step (lag 1) beats default lag 3 by 1.3x–2.6x with <0.04% physics error



Study S03: Layered Options & Controller Gains

Compounding wins: pairing lag 1 with tuned PID controller gains yields up to 26x transient speedup and 2.5x full-run win

